

Introduction

Can low-cost CNCs be made secure and intelligent?

- Traditional CNCs are closed-architecture and limit real-time monitoring or control
- We convert a Langmuir MR-1 into a flexible, cyber-secure, and data-rich closed-architecture system
- We retrofit a Langmuir MR-1 with multimodal sensors such as Accelerometers, AE sensors, etc.
- Our testbed supports real-time command execution, sensor logging, and on-device anomaly detection

Background

- Most CNC controllers are closed-architecture and restrict access to firmware modifications
- Cyberattacks targeting physical CNC systems remain critically understudied
- Real-time data is critical for adaptive control and fault detection
- Open systems enable reproducible, custom research in smart and secure manufacturing

Methods

- Modified GRBL firmware integrated with Langmuir MR-1 CNC machine
- MATLAB used for serial CLI communication at 115,200 baud rate
- Streamed G-code directly from a local computer for precise, repeatable motion control

Sensing & Data Collection

- CNC position logged in real time at 15-20 Hz
- AprilTags + Camera used for tool tracking
- Multimodal sensors (Accelerometer, AE, and Camera) instrumented for side-channel monitoring
- All sensor streams time-synchronized and fused for downstream analysis

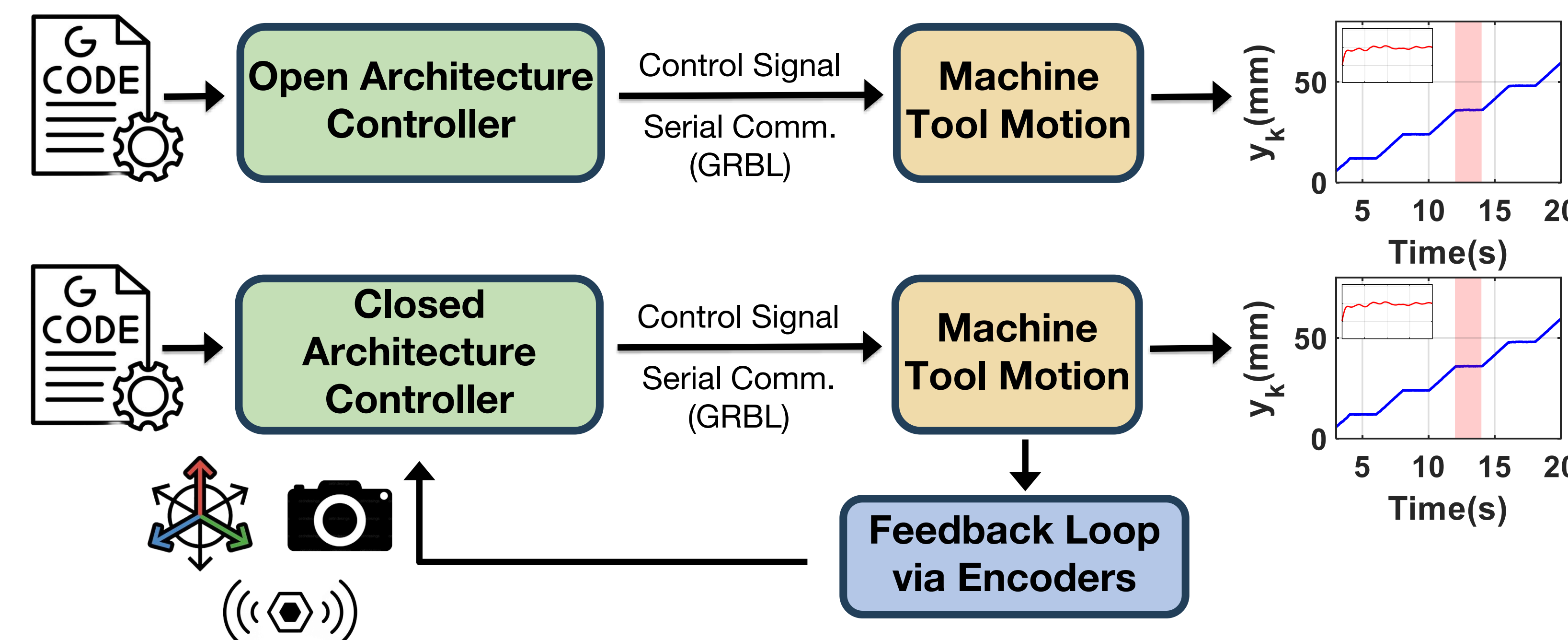


Figure 1. Open-loop versus closed-loop machine tool motion schematic.

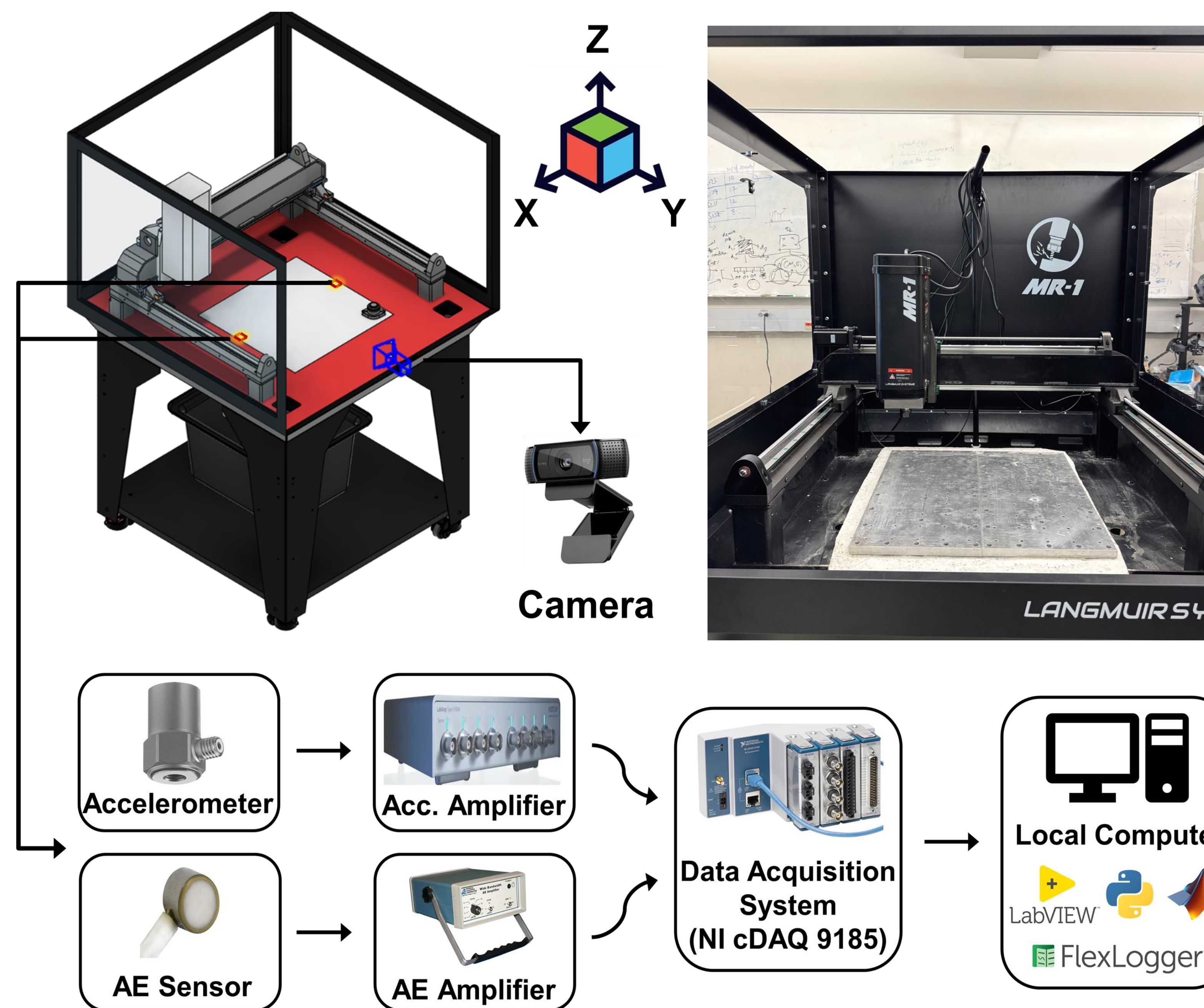


Figure 2. Plug-and-play end-to-end data pipeline.

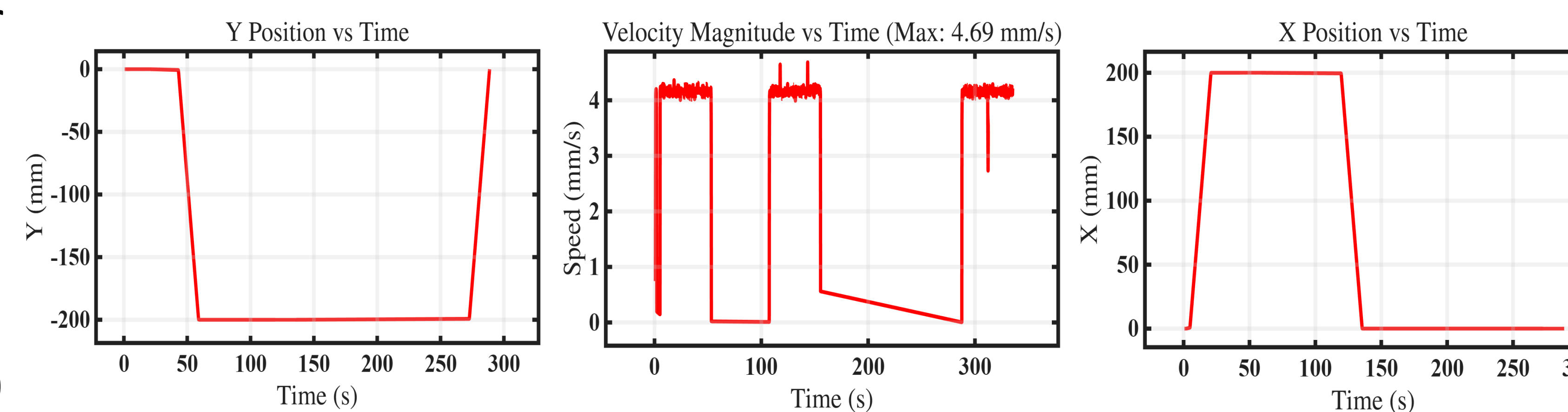


Figure 3. X and Y positions with corresponding velocity magnitude during a CNC square path. Motion shows clear directional changes and consistent speed along each segment.

Results

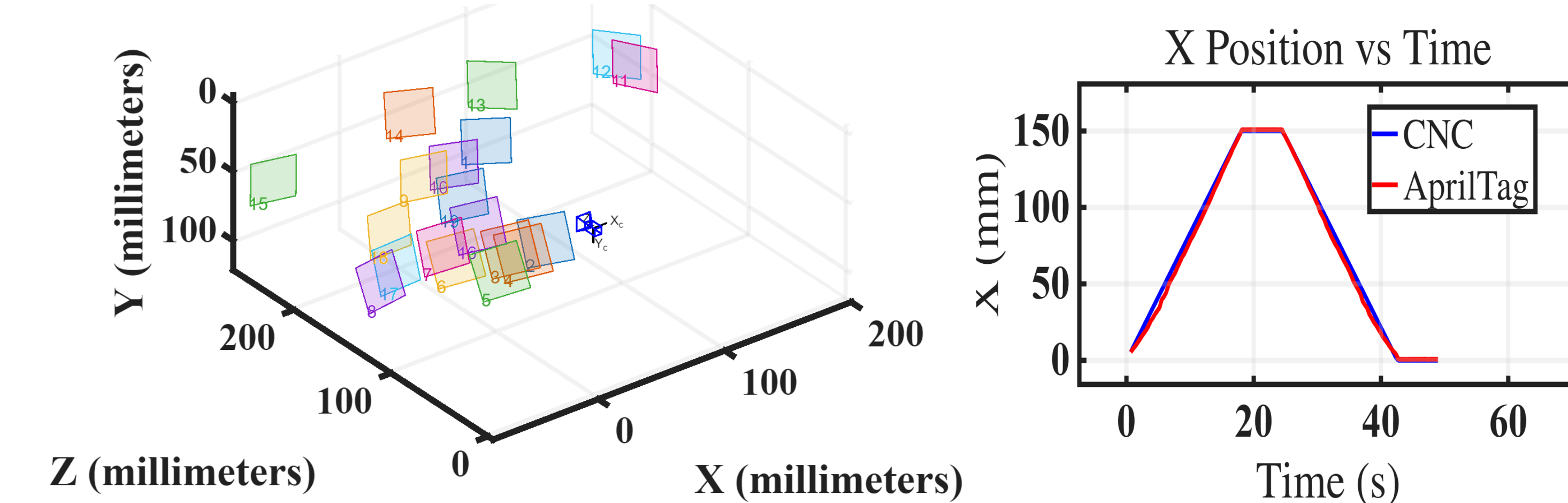


Figure 4. Camera Calibrator App

Figure 5. CNC Vs. AprilTag Tracking on X-Axis

Outcomes

- Synchronized multimodal sensor data with sub-millimeter agreement between CNC and visual tracking (error $< \pm 1$ mm)
- Accelerometer and AE sensors help identify the system dynamics and potential anomalous behavior
- Setup enables future research in real-time anomaly detection and closed-loop adaptive control

Future Work

- Extend closed-loop control using setpoint corrections from multimodal sensor feedback
- Develop and inject controlled CLI-based cyberattack vectors (e.g., injection, replay, and flip attacks)
- Implement ML-based anomaly detection using multimodal sensor fusion monitoring
- Implement Reinforcement Learning (RL) based recovery and mitigation

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